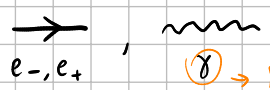
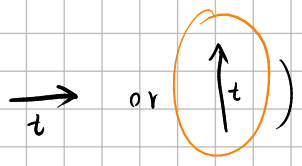
Feynman rules for QED

interaction is $e\,\bar{\Psi}\gamma^\mu A_\mu \Psi$

$\bar{\Psi}$ → positron, A_μ → photon, Ψ → electron

diagrams have as ingredients: 

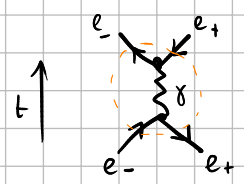
vertices

once you chose time direction (major conventions are 

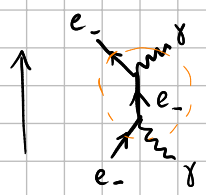
)

if the fermion arrow is in same direction as time we say it's e_-, otherwise it's e_+

examples:



Bhabha scattering



Compton scattering

electrons have spin and helicity
photons have polarization
} → only external ones

each external leg also carries information about these things

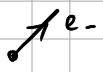
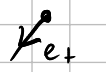
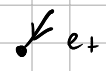
Feynman rules (momentum space)

- each vertex has a factor of $ie\gamma^\mu$
- each internal photon between vertices $ie\gamma^\mu$, $ie\gamma^\nu$ carries a photon propagator

  $\frac{-i\eta_{\mu\nu}}{k^2}$ (k 4-momentum of photon)

- each internal fermion carries a fermion propagator: $\frac{i(\not{p}+m)}{p^2-m^2}$ (p 4-momentum of electron / positron)
- each vertex has integration over all momenta of internal legs and a δ of sum of momenta (conservation of momentum)
- external legs carry extra things

↳

- e^- in a vertex: carries a factor of u_s (spinor)
- e^- out a vertex:  carries a factor of $\bar{u}_s = u_s^* \gamma^0$
- e_+ in a vertex:  carries a factor of $\bar{v}_s$
- e_+ out a vertex:  carries a factor of v_s
- γ in a vertex carries a factor of ε_μ^s (polarization)
- γ out a vertex carries a factor of $(\varepsilon_\mu^s)^*$ (polarization)

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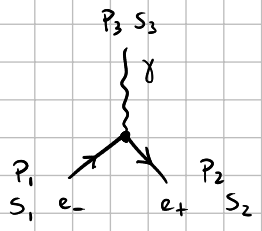
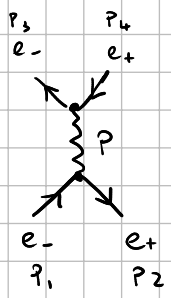
ex



$$\bar{v}_{s_2}(p_2)\; i e \gamma^\mu \; (\varepsilon_\mu^{s_3})^*(p_3)\; u_{s_1}(p_1)$$



$$\left[\bar{v}_{s_2}(p_2)\, i e \gamma^\mu u_{s_1}(p_1)\right]\left(\frac{-i\eta_{\mu\nu}}{(p_1+p_2)^2}\right)\left[\bar{u}_{s_3}(p_3)\, i e \gamma^\nu v_{s_4}(p_4)\right]$$

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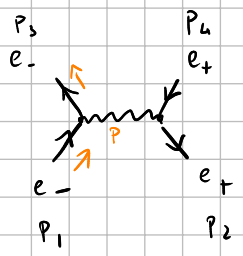
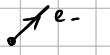
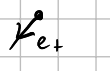
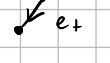
how to figure out ε_μ: $\varepsilon^\mu(p)\, p_\mu = 0$

$$\sum_s \varepsilon_\mu^s(p)\, \varepsilon_\nu^s(p) = -\eta_{\mu\nu}$$

polarization

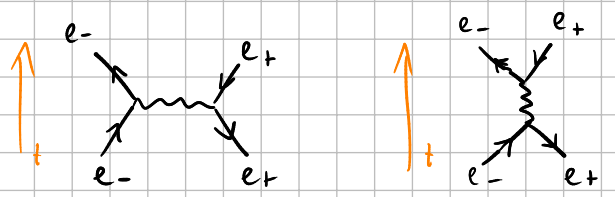


$$\left[\bar{u}_{s_3}\, i e \gamma^\mu u_{s_1}\right] \frac{-i\eta_{\mu\nu}}{(p_1-p_3)^2} \left[\bar{v}_{s_2}\, i e \gamma^\nu v_{s_4}\right]$$

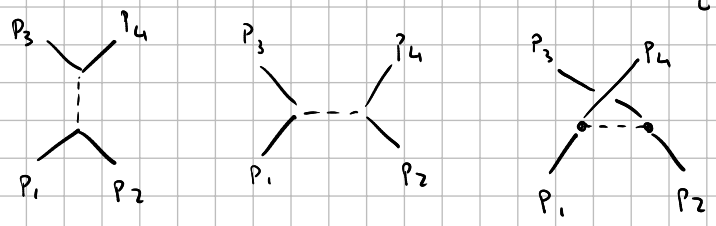
- e^- in a vertex: carries a factor of u_s (spinor)
- e^- out a vertex:  carries a factor of $\bar{u}_s = u_s^* \gamma^0$
- e_+ in a vertex:  carries a factor of $\bar{v}_s$
- e_+ out a vertex:  carries a factor of v_s

Bhabha scattering

look at 2 possible diagrams



t channel s channel



s-channel t-channel u-channel

$\mathcal{M}$ amplitude of scattering: $\mathcal{M} = \mathcal{M}_t - \mathcal{M}_s$

spinors